Customer Segmentation — Sephora

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1. Data Understanding

1.1 Grain of the Table

The dataset contains **399,997 transaction lines** spanning January–December 2025. Each row represents *one product line within one ticket*: a single SKU (`materialCode`) purchased by a customer (`anonymized_card_code`) inside a ticket (`anonymized_Ticket_ID`). The data covers **64,469 unique customers** and **184,282 unique tickets**, yielding on average **3.0 tickets/customer** (median = 2) and **2.2 lines/ticket** (median = 2). Transactions span three countries (FR 98.6%, LU 0.96%, MC 0.46%) and two channels (store 86%, estore 14%).

Aggregation plan. Because the unit of analysis for segmentation is the *customer*, we will collapse these rows into a single feature vector per `anonymized_card_code`: RFM metrics, channel mix, product-axis affinity shares, discount sensitivity, and demographic proxies — yielding a $64{,}469 \times p$ feature matrix ready for clustering.

1.2 Data Quality Findings

Finding	Impact & Resolution
Typo: `MAEK UP` (115,679 rows) vs. `MAKE UP` (77,411)	Two labels for the same axis. After merge: Make Up becomes the **#1 axis by volume** (193,090 rows, 35.9% of revenue). Must fix before any axis-based feature.
Missing first-purchase fields (74% null across 10 columns)	Only ∼26% of customers have first-purchase details. We will create a binary flag `has_first_purchase` and use these fields only as secondary enrichment, never as core clustering features.
Age = 0 (8.6% of rows, ∼5,600 customers)	Encode as `NaN`; rely on `age_category/age_generation` when available. 28–29% of rows lack age categories — age will be treated as supplementary, not primary.
Negative sales (5,210 rows, 1.3%) & zero sales (8,893 rows)	Returns and samples. We keep them to reflect net customer value: a customer who returns a lot is behaviorally different from one who doesn't.
Gender: 99,999 = undeclared (23 rows / 6 customers)	Negligible. Recode to "Unknown". 88% of customers are Women, 12% Men.

1.3 Key Numbers

- **Revenue**: €12.77M (net incl. VAT)
- **Avg. customer LTV**: €198 (med. €108)
- **Avg. basket**: €69.5 (med. €54)
- **Discount rate**: 13.1%
- **Recency**: 111d mean (med. 68d)
- **Loyalty**: BRONZE 63.6%, SILVER 32.1%, GOLD 4.3%
- **Store-only**: 80.4%; Omni: 12.5%; Estore: 7.1%
- **Top axis**: Fragrance 37.8%, Make Up 35.9%
- **Market**: Selective 48.4%, Exclusive 36.0%

2. Angle: Behavioral Segmentation via Unsupervised Clustering

Why. Sephora's existing RFM segments (9 IDs) collapse 64k customers into broad buckets that conflate channel behavior, product affinity, and value tiers. Our hypothesis is that introducing *multi-dimensional* clustering — combining RFM, channel mix, product-axis shares, discount sensitivity, and demographic signals — will reveal **latent personas** that are more actionable for marketing than RFM alone.

What. We target 10–30 clusters evaluated by Silhouette Score (≥ 0.35), Davies–Bouldin Index, and Calinski–Harabasz Index, profiled via KPI delta tables.

2.1 Exploratory Evidence

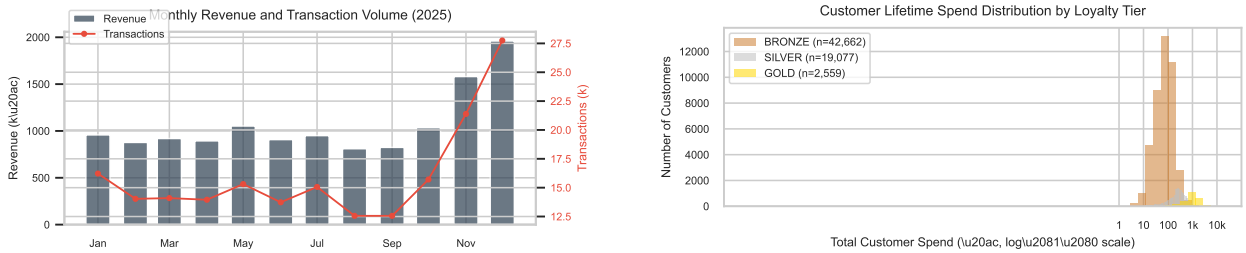

Figure 1: Left: Strong Nov–Dec seasonality (27.7% of revenue) — seasonal buyers vs. year-round loyalists. **Right**: Log-normal spend; BRONZE hides sub-segments.

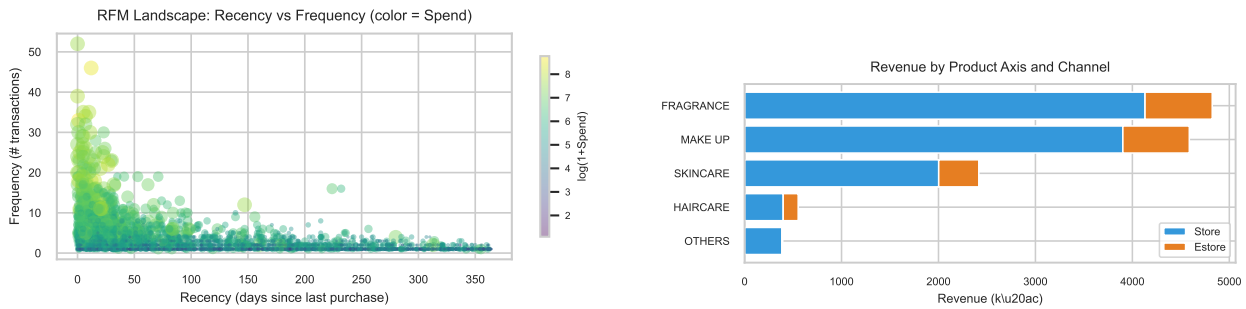

Figure 2: Left: Dense low-freq/recent mass + sparse high-freq loyalist tail. **Right**: Fragrance is in-store dominated; Make Up drives estore — channel×axis clusters expected.

2.2 Initial Hypotheses

1. **BRONZE is not monolithic**: 63.6% of customers — likely 3–4 sub-segments (one-time gifters, digital explorers, bargain hunters, dormant). Log-spend (Fig. 1R) confirms high within-tier variance.
2. **Channel×Axis drives personas**: 80% store-only but omnichannel 12.5% over-indexes on value. Fragrance is in-store; Make Up drives estore (Fig. 2R).
3. **Seasonality defines behavior**: Nov–Dec spike (+80% vs. Aug–Sep) suggests a "gifter" cluster active only in Q4.
4. **RFM→15–25 personas** once we add channel, product, and discount dimensions.

3. What Comes Next

Week-ahead plan (Sprint 2):
(1) Feature engineering — customer-level RFM + channel share + axis shares + discount rate + demographics (∼15–20 features).
(2) Preprocessing — StandardScaler, PCA (≥85% var.), UMAP 2D.
(3) Clustering — K-Means, Agglomerative, DBSCAN, GMM; $k \in [5, 30]$; select via Silhouette + Davies–Bouldin.
(4) Profiling — KPI delta tables + Kruskal–Wallis significance.
(5) Persona cards — marketing names + campaign recommendations.

Key unknowns: (a) Include customers with <2 txn or not; (b) PCA vs. raw features for cluster stability; (c) Handling of 74%-missing first-purchase block.

Conversation starter — all numbers from raw dataset, subject to refinement.